SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	1 July 2026
Team ID	xxxxxx
Project Name	A Comphrensive Measure of Well Being
Maximum Marks	3 Marks

Project Executable Files

This section contains all the executable files required for the **A Comprehensive Measure of Well Being** project. It includes the Flask application source code, trained machine learning model (`model.pkl`), HTML templates, static files, and the required dependency file. All project files are properly organized with clear run instructions so that the evaluator can easily execute the application locally and generate Human Development Index (HDI) predictions**.**

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/ No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example similar)	No
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA

9	Test files and test results	No
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
Human_Development_Index/
|
├── app.py
|   └── Main Flask application
|
├── generate_dataset.py
|   └── Generates the synthetic HDI dataset
|
├── train_model.py
|   └── Trains and evaluates the Linear Regression model
|
├── hdi_dataset.csv
|   └── Generated synthetic training dataset
|
├── model.pkl
|   └── Saved trained Linear Regression model
|
├── requirements.txt
|   └── Python package dependencies
|
├── templates/
|   |
|   ├── index.html
|   |   └── HDI input form / home page
|   |
|   └── result.html
|       └── Prediction result and HDI category page
|
└── static/
    |
    └── css/
        |
        └── style.css
            └── Web application styling
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://127.0.0.1:5000
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Render
Repository Link	https://github.com/yourusername/hdi-prediction
Demo Video Link	https://1drv.ms/v/c/26cc38ef316efa52/IQAPlqVlt3fbSrgTT-pc_Ov1Acj6UP-bniihJwoOeRx6z5E?e=PMVgkD

Step 4: Run Instructions

1. Download or clone the project source code and open the `Human_Development_Index` project folder in Visual Studio Code.

2. Open a terminal in the project root directory.

3. Install all required Python dependencies using the command `pip install -r requirements.txt.

4. Run `python generate_dataset.py` to generate the synthetic Human Development Index dataset. The generated dataset is saved as `hdi_dataset.csv.

5. Run `python train_model.py` to load the dataset, split it into training and testing sets, train the Linear Regression model, evaluate model performance, and save the trained model as `model.pkl.

6. Run `python app.py` to start the Flask web application.

7. Open a web browser and navigate to `http://127.0.0.1:5000.

8. Enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income in the input form.

9. Click the `Predict HDI` button. The application sends the input data to the Flask `/predict` route, converts the values into a NumPy array, and uses the trained Linear Regression model to generate the predicted HDI value.

10. The application displays the predicted HDI value and corresponding development category: Low, Medium, High, or Very High Human Development.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the trained machine learning model	Improve the model using more training data
2	The application accepts only four input parameters	Additional HDI indicators can be incorporated in future versions
3	The application currently runs on a local Flask server	It can be deployed on cloud platforms such as Render or Python Anywhere